

Experiment-07: Verification of Thevenin's, Norton's and Maximum Power Transfer Theorem

Name: Raihan Ahammed

ID: 2532600043

Group: 03

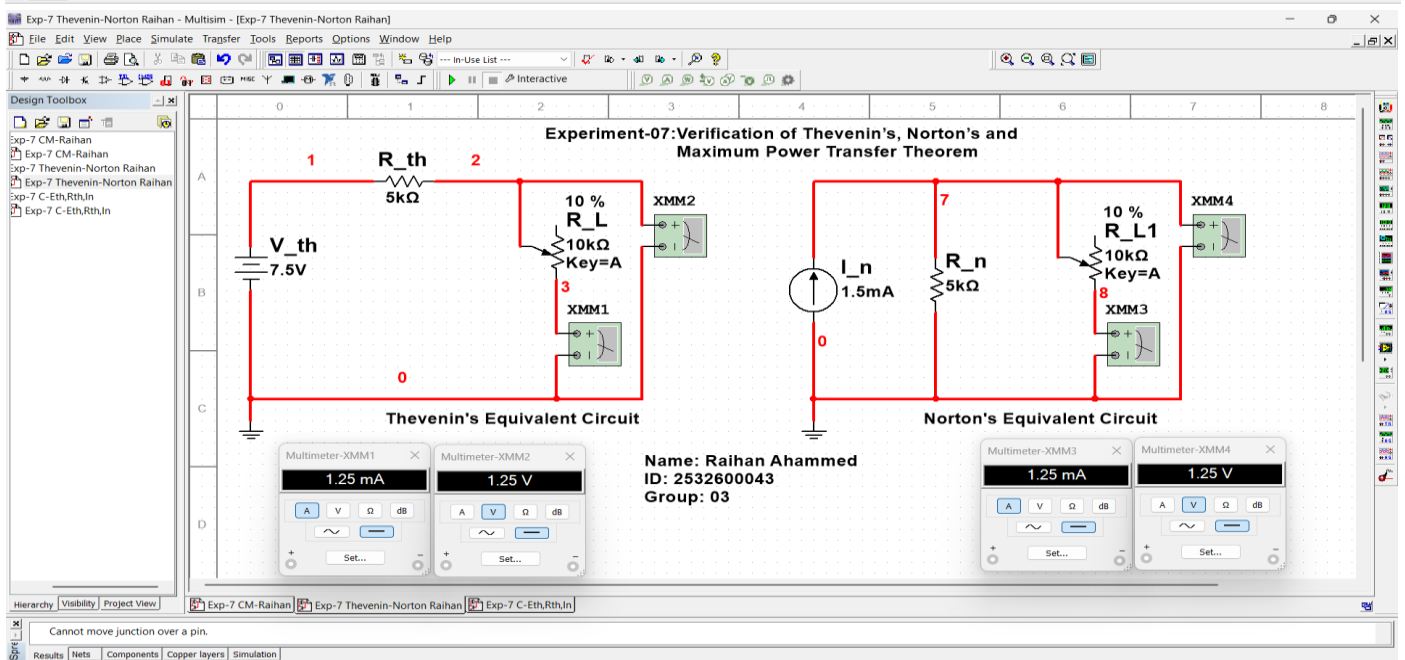
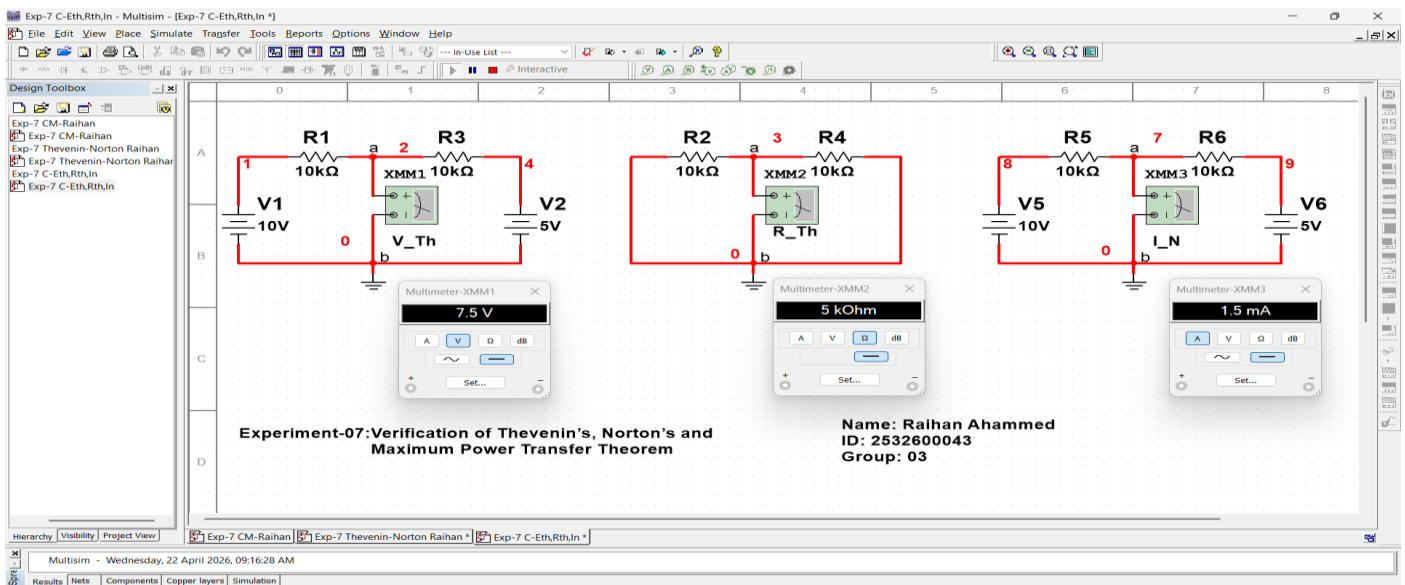
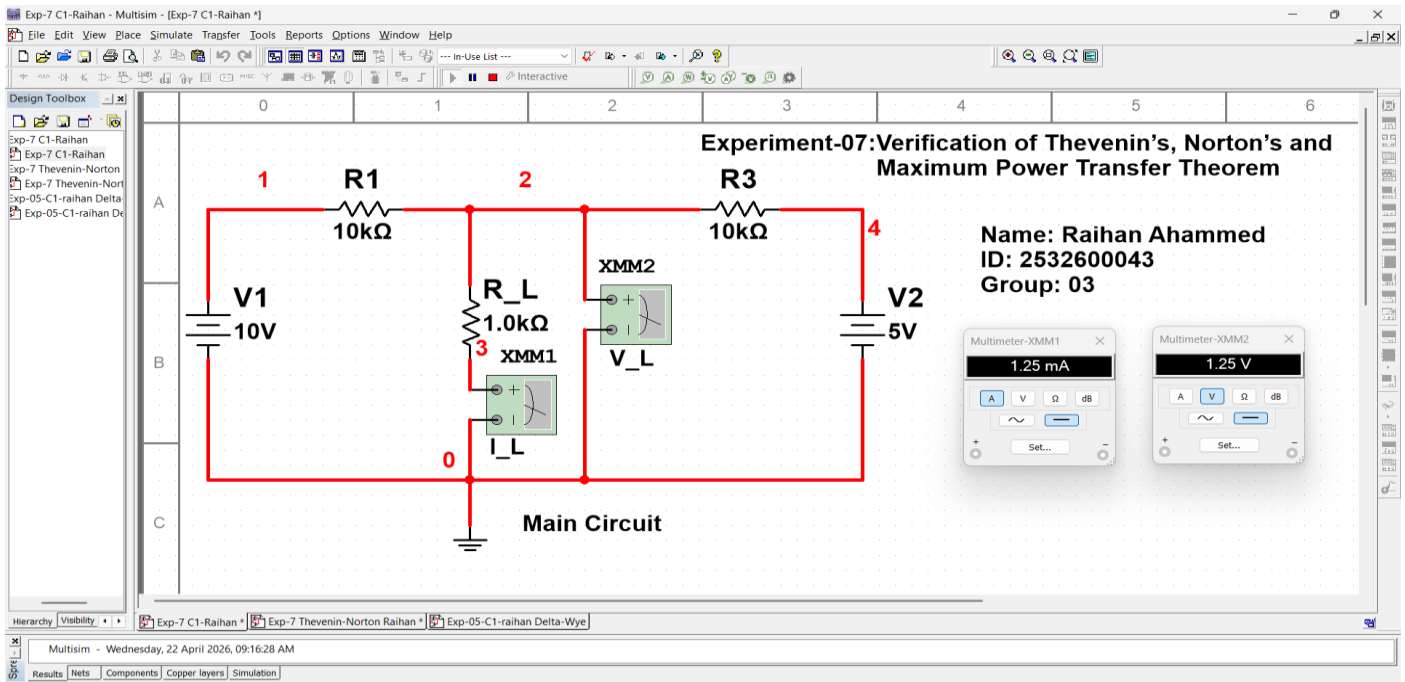


Table-1:

Measurement	Measured (DMM)
V_{TH}	7.5 V
R_{TH}	5 k Ω
I_N	1.5 mA

Table 2:

Value	Main Circuit	Thevenin's Eq Circuit	Norton's Eq Circuit
V_L	1.25V	1.25V	1.25V
I_L	1.25mA	1.25mA	1.25mA

Table 3:

R_L (k Ω)	V_L (V)	P_L (mW) V_L^2 / R_L
1.0 k Ω	1.25 V	1.56 mW
2.0 k Ω	2.14 V	2.30 mW
3.0 k Ω	2.81 V	2.64 mW
4.0 k Ω	3.33 V	2.78 mW
5.0 k Ω	3.75 V	2.81 mW
6.0 k Ω	4.09 V	2.79 mW
7.0 k Ω	4.37 V	2.73 mW
8.0 k Ω	4.62 V	2.66 mW
9.0 k Ω	4.82 V	2.58 mW
10 k Ω	5.00 V	2.50 mW